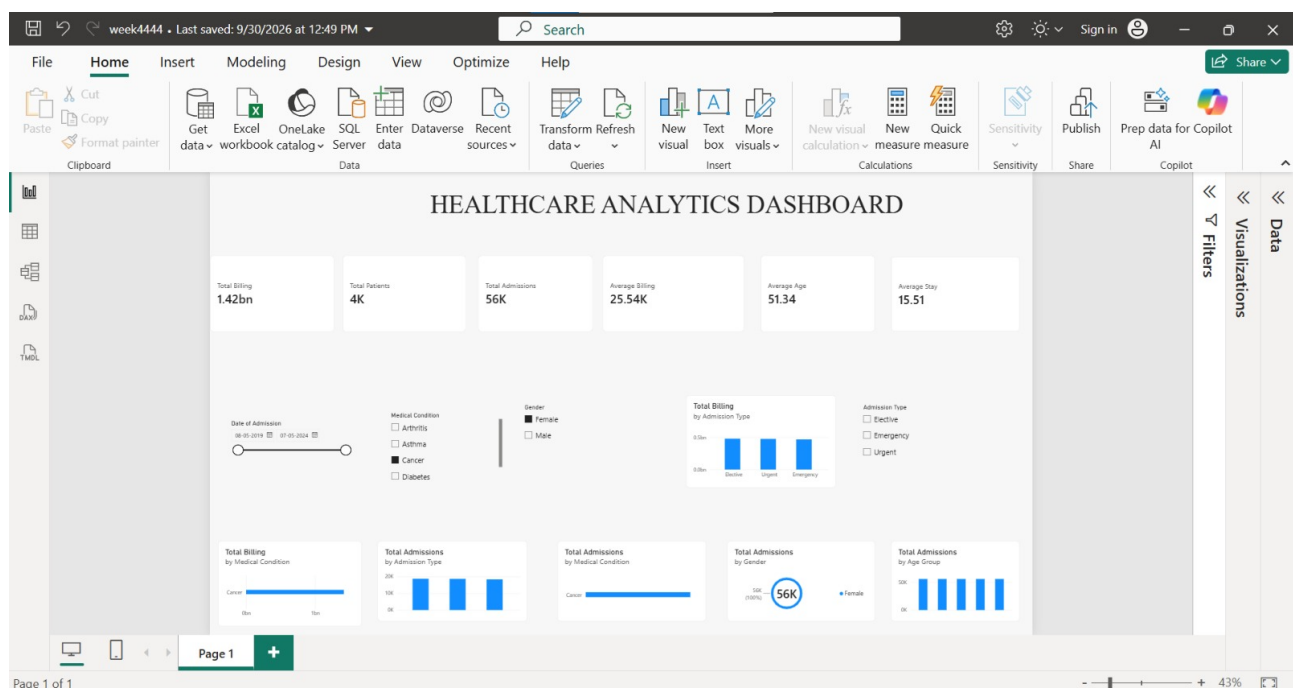


HEALTHCARE ANALYTICS DASHBOARD

Power BI – Week 5 Project Report



Final interactive dashboard developed in Power BI

Project	Healthcare Analytics Dashboard
Tool	Microsoft Power BI Desktop
Data Source	Healthcare dataset
Project Stage	Week 5
Main Focus	Data modelling, measures, filters and dashboard visualisation

Prepared as a professional documentation report for Week 5 Power BI work.

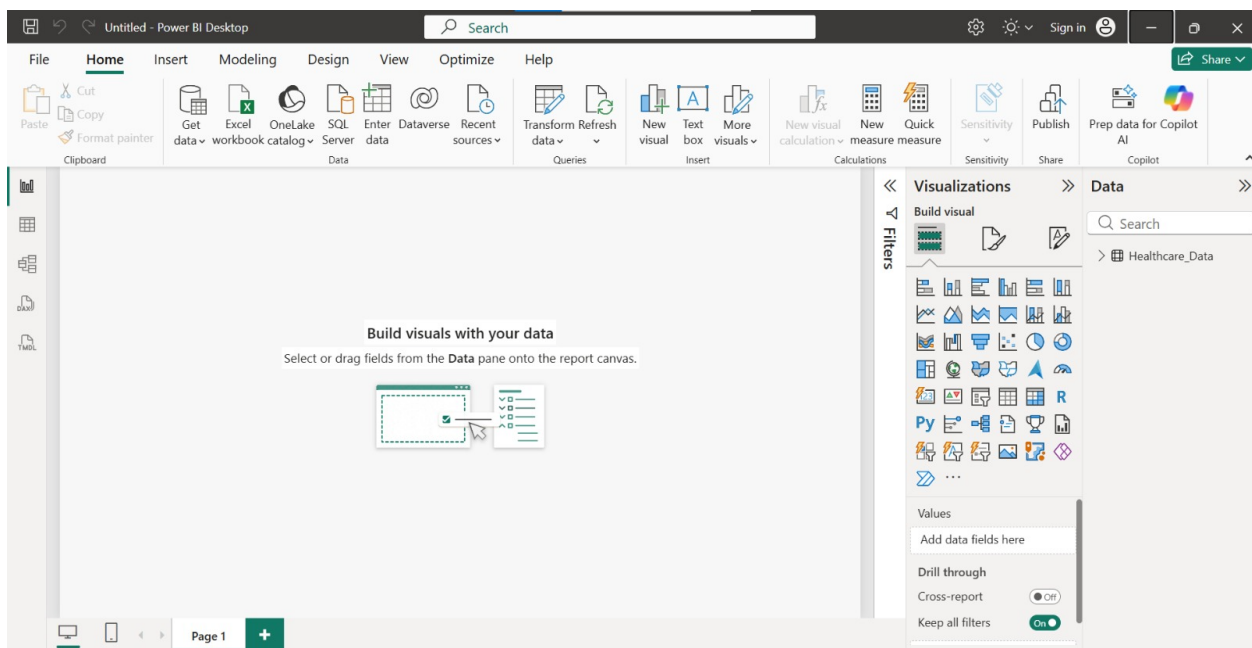
1. Project Overview

The objective of this project is to transform healthcare records into an interactive analytical dashboard. The report combines patient, admission and billing information so that users can explore key metrics and compare healthcare activity using filters and visual charts.

The Week 5 implementation focuses on building a structured Power BI model, creating calculated measures, and presenting the results through a clean dashboard layout.

Key analytical areas

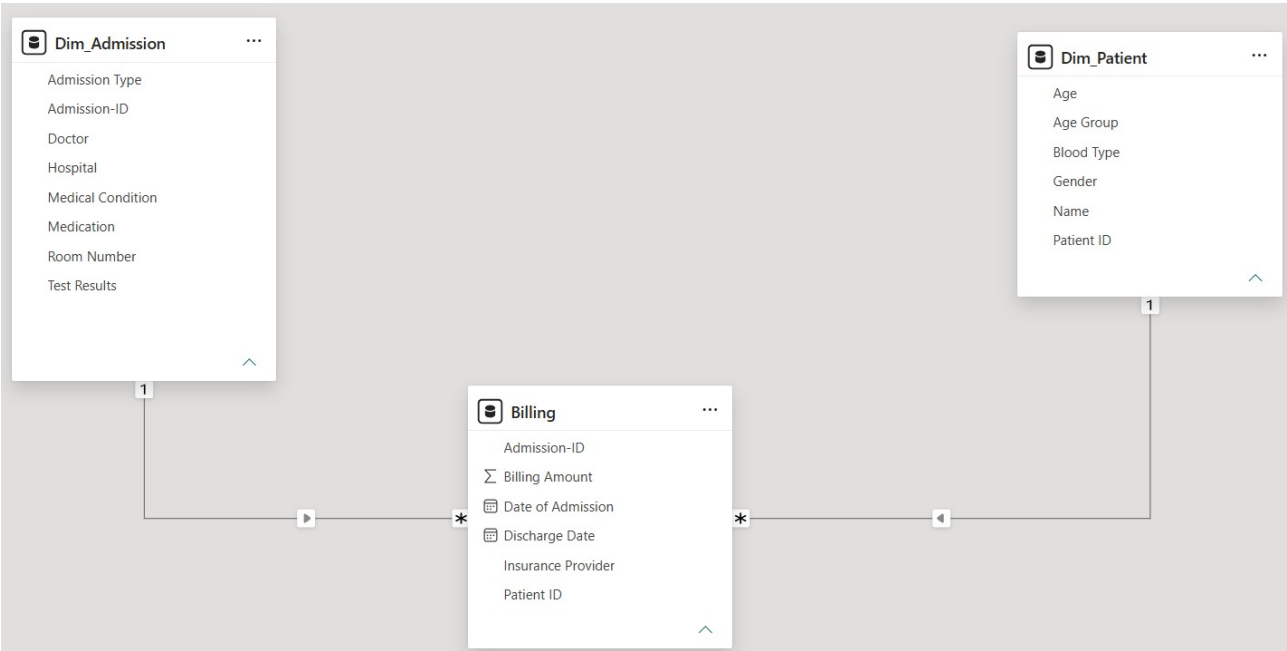
- Patient volume and demographic information
- Admission types and admission counts
- Billing amount and average billing
- Medical-condition based analysis
- Gender and age-group analysis
- Date-based filtering and interactive exploration



Initial Power BI report canvas used before the dashboard was assembled.

2. Data Model and Relationships

The Power BI model separates the healthcare information into logical tables. The screenshots show a main healthcare dataset together with dimension-style tables for patient and admission information and a billing table. Relationships were established so that visuals can be filtered consistently across the model.



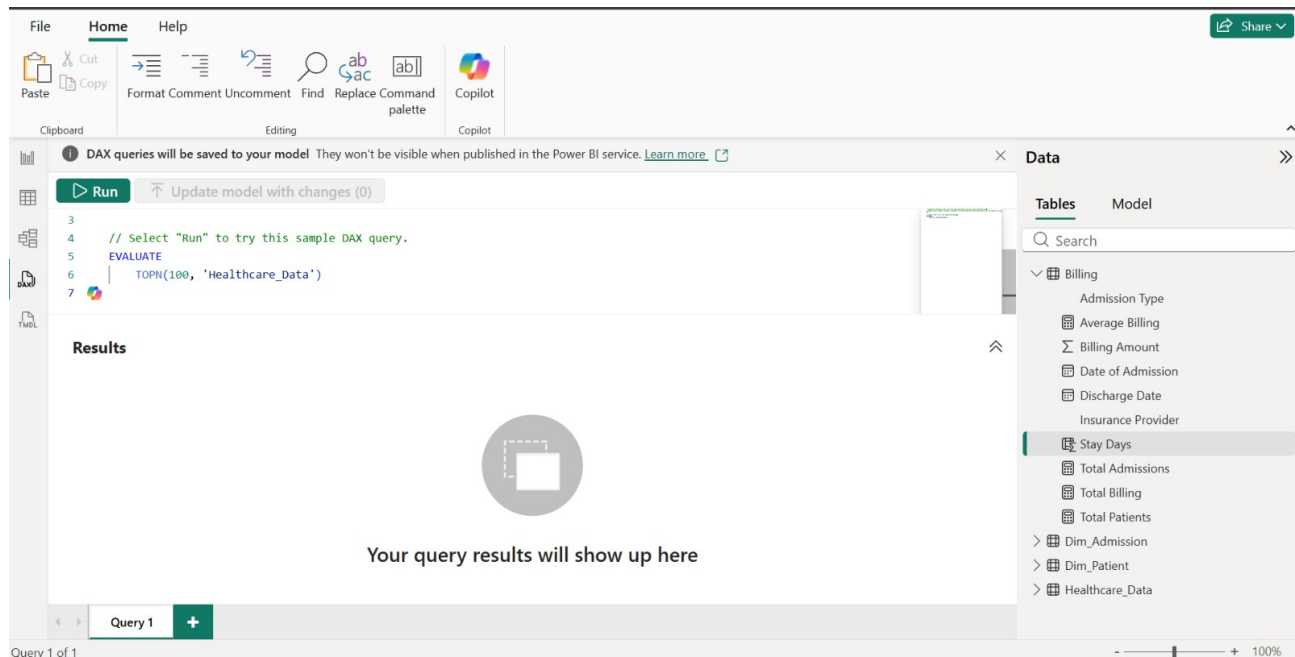
Relationship/model view captured during Week 5 development.

Model components visible in the project

Table	Purpose
Healthcare_Data	Main healthcare records and source-level attributes
Dim_Patient	Patient-related attributes such as age, age group, gender and blood type
Dim_Admission	Admission-related attributes such as doctor, hospital, medical condition and medication
Billing	Billing amount, admission/discharge dates and billing-related measures

3. Measures and Calculations

Measures were created in Power BI to calculate summary values dynamically. The project includes measures such as Total Billing, Total Admissions, Total Patients, Average Billing and Average Stay. These measures are then reused in cards and charts.



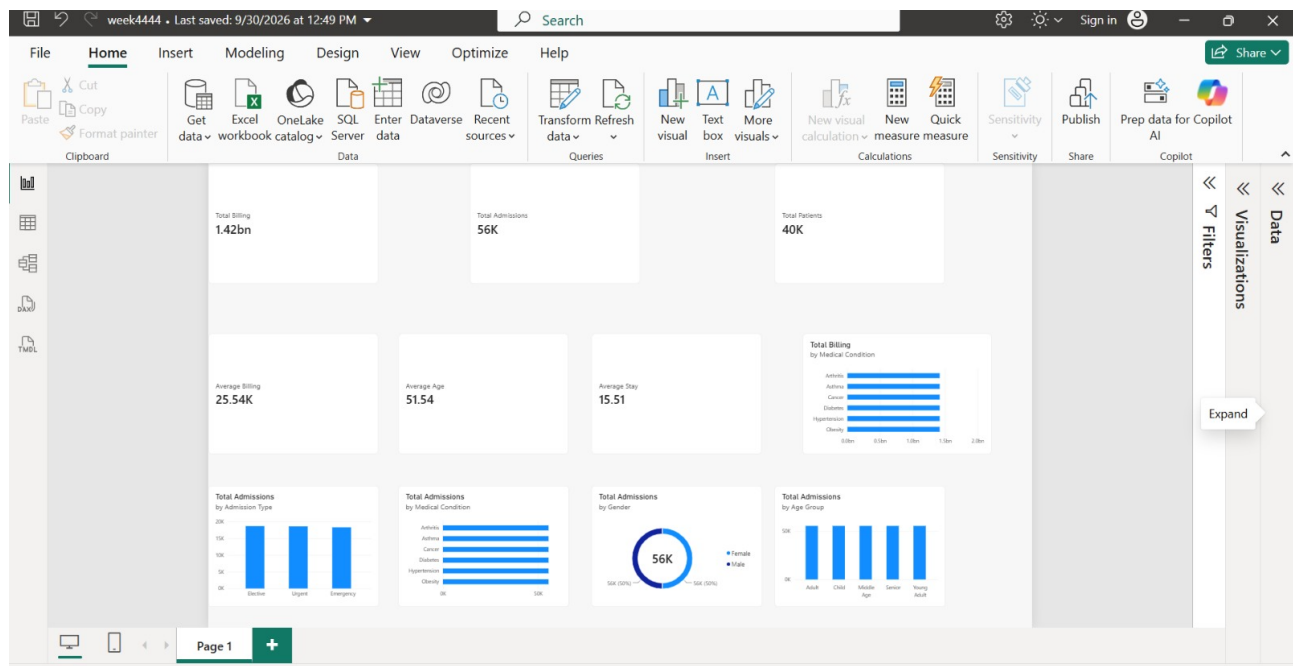
Power BI DAX query/model area used while validating the data model and measures.

Main measures used in the dashboard

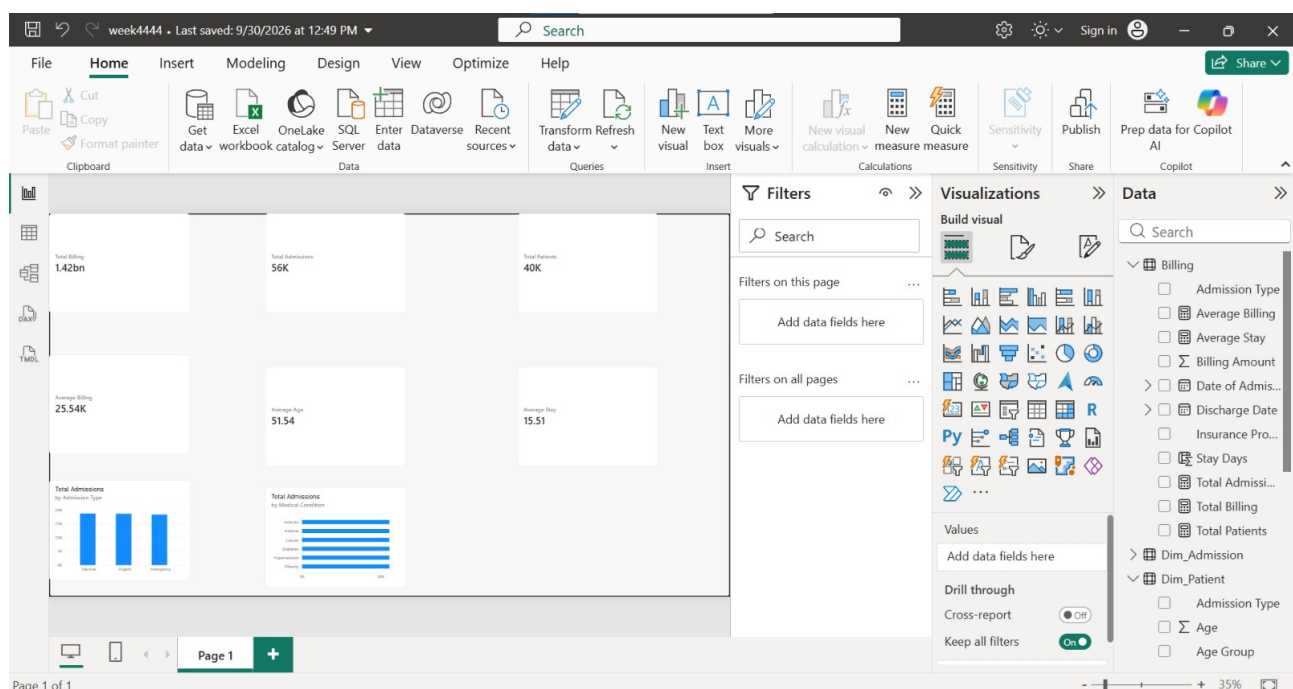
Measure	Dashboard purpose
Total Billing	Displays the total billing amount
Total Patients	Shows the number of patients represented in the analysis
Total Admissions	Shows the number of admission records
Average Billing	Shows the average billing amount
Average Age	Shows the average patient age
Average Stay	Shows the average stay duration

4. Dashboard Development

The dashboard was developed progressively. KPI cards were placed at the top for quick summary, followed by interactive slicers and analytical charts. This structure keeps the most important information visible while allowing users to drill into specific categories.



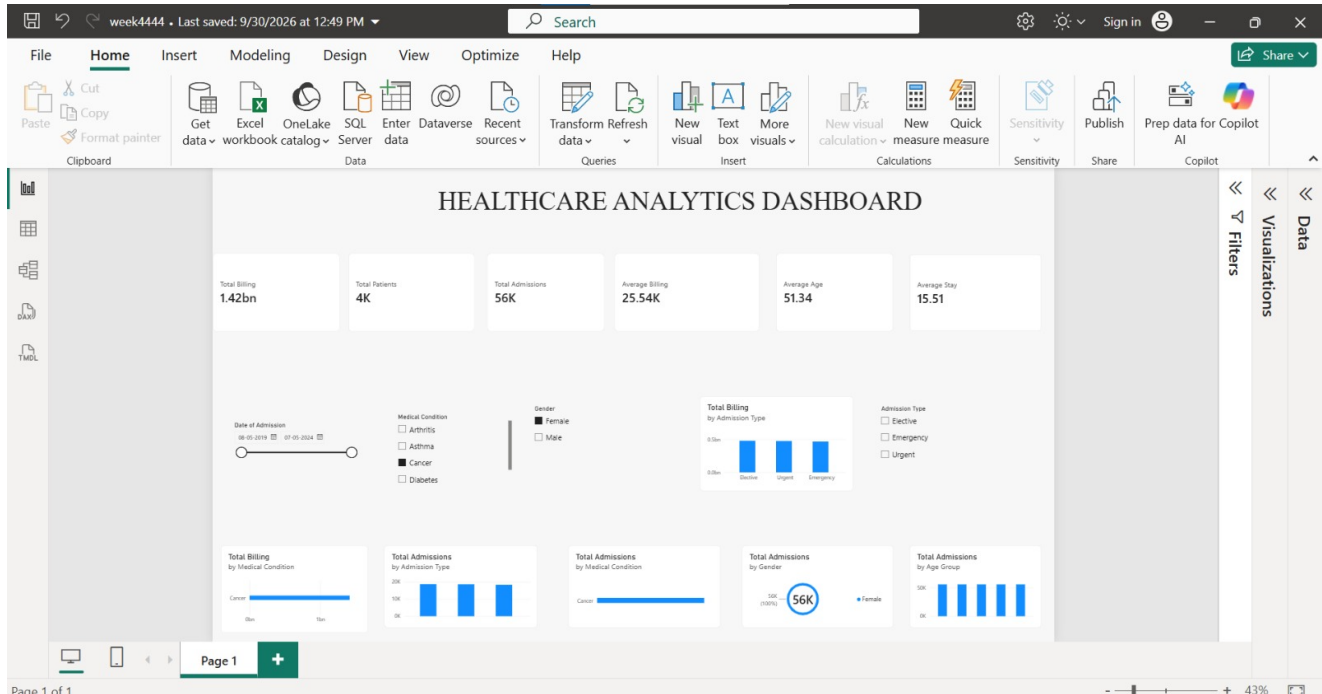
Intermediate dashboard layout showing KPI cards and analytical visuals.



Further refinement of the chart layout and spacing.

5. Final Dashboard

The final dashboard combines KPI cards, slicers and charts into one interactive report page. The visible controls include date selection, medical condition, gender and admission type. The charts cover billing by medical condition, admissions by admission type, admissions by medical condition, gender distribution and age-group distribution.



Final Healthcare Analytics Dashboard – Week 5.

Dashboard components

- KPI cards: Total Billing, Total Patients, Total Admissions, Average Billing, Average Age and Average Stay.
- Date slicer: enables analysis across a selected admission-date range.
- Category slicers: medical condition, gender and admission type.
- Billing analysis: compares billing across medical conditions.
- Admission analysis: compares admissions by admission type and medical condition.
- Demographic analysis: presents gender and age-group distributions.

6. Testing and Validation

The dashboard was checked visually after adding measures and charts. The model view was reviewed to confirm that the required tables were present, while the report page was checked to ensure that slicers and visuals respond to selections.

Validation checklist

Check	Status
Tables available in the Power BI model	Completed
Relationships visible in Model view	Completed
Measures created and available for visuals	Completed
KPI cards populated	Completed
Charts and slicers placed on the report page	Completed
Final dashboard arranged as a single analytical page	Completed

7. Conclusion

The Week 5 project converts healthcare records into a structured and interactive Power BI dashboard. The completed report demonstrates data modelling, DAX-based measures, KPI reporting, slicers and multiple analytical visualisations. The resulting page provides a consolidated view of patient, admission and billing information and can serve as the foundation for the next stage of the project.

Recommended continuation for Week 6

Continue from the completed Power BI file rather than rebuilding the model. Reuse the existing tables, relationships, measures and dashboard page, then add the Week 6 requirements on top of this foundation.